

BHARATIYA VIDYA BHAVAN'S SARDAR PATEL INSTITUTE OF TECHNOLOGY

(AUTONOMOUS INSTITUTE AFFILIATED TO UNIVERSITY OF MUMBAI)

ACADEMIC YEAR: 2026 - 2027 SEM: III

Department of Computer Science and Engineering

Database Management Systems

Mini-Project Abstract

Topic: College Clubs and Committees Event Management System

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1. Abstract

Colleges organize various technical, cultural, sports, and extracurricular activities through student-run clubs and committees. However, managing memberships, event proposals, faculty approvals, venue bookings, and registrations through separate spreadsheets and manual processes often lead to data duplication, scheduling conflicts, and administrative delays. The proposed College Clubs and Committees Event Management System addresses these challenges by providing a centralized relational database for efficient management of college activities.

The primary objectives of the system are to maintain student and faculty records, manage club memberships within a three-tier hierarchy (Supercore, Core, and Subcommittee), streamline event approvals, prevent venue conflicts and duplicate registrations, track attendance, and collect participant feedback. It also supports official announcements and role-based access for students, faculty, and administrators.

The system allows authorized members to propose events, assign venues, and specify participant capacities. Faculty advisors or administrators review event proposals before registrations begin. Students can register for approved events, while attendance records and feedback are maintained after event completion. Database constraints ensure data consistency, prevent duplicate entries, and maintain referential integrity.

The database has been designed and implemented using MySQL 8.0+, with nine interconnected tables representing the major entities. It incorporates primary keys, foreign keys, unique constraints, and validation rules. The schema is normalized up to Third Normal Form (3NF) to minimize redundancy and maintain data integrity. This database implementation establishes the foundation for future frontend and backend development, ultimately supporting a complete centralized college event management system.

2. Proposed Entities

Entity	Primary Key	Foreign Keys	Other Attributes
Roles	role_id	None	role_name
Users	user_id	role_id → Roles	full_name, email, roll_no, department
Organizations	organization_id	faculty_advisor_id → Users	org_name, description
Memberships	membership_id	organization_id → Organizations; user_id → Users	member_role
Venues	venue_id	None	venue_name, capacity
Events	event_id	organization_id → Organizations; venue_id → Venues; created_by → Users; approved_by → Users	event_name, description, event_date, max_participants, approval_status, event_status
Registrations	registration_id	event_id → Events; user_id → Users	participation_status
Feedback	feedback_id	registration_id → Registrations	rating, comments
Announcements	announcement_id	organization_id → Organizations; created_by → Users	title, description

3. Constraints Enforced and Relationship Cardinalities

Type	Applied on
NOT NULL	Names, emails, department, role_name, org_name, venue_name, capacity, event_name, event_date, max_participants, approval_status, event_status, participation_status, rating, title, description
UNIQUE	Emails (users), roll_no (users), role_name (roles), org_name (organizations), venue_name (venues), Memberships (organization_id, user_id), Registrations (event_id, user_id), Feedback.registration_id
CHECK	capacity > 0 (venues); max_participants > 0 (events); rating BETWEEN 1 AND 5 (feedback); valid values for member_role, approval_status, event_status, and participation_status
DEFAULT	Registration participation_status = 'Registered'; approval_status = 'Pending'; event_status = 'Upcoming'; member_role = 'Member'
Referential integrity	ON UPDATE CASCADE across all foreign keys; ON DELETE CASCADE for dependent records (Memberships, Registrations, Feedback, Announcements); ON DELETE SET NULL for optional references (faculty_advisor_id, approved_by)
Cardinality	All relationships are 1 : N (Users–Memberships, Organizations–Events, Venues–Events, Events–Registrations, etc.), except Registrations–Feedback, which is 1 : 1

4. Sample Tables

roles

	role_id	role_name
▶ 1	1	Student
2	2	Faculty
3	3	Administrator
*	NULL	NULL

users

	user_id	role_id	full_name	email	roll_no	department
▶ 1	1	3	Dr. B. N. Chaudhari	principal@spit.ac.in	NULL	Administration
2	2	2	Dr. Anand Godbole	anand.godbole@spit.ac.in	NULL	Computer Engineering
3	3	1	Mihir Pukale	mihir.pukale@spit.ac.in	2023300001	Computer Engineering
4	4	1	Rushikesh Patil	rushikesh.patil@spit.ac.in	2025800093	Computer Engineering
5	5	1	Vighnesh Pawar	vighnesh.pawar@spit.ac.in	2025800098	Computer Engineering
6	6	1	Arjun Thakur	arjun.thakur@spit.ac.in	2023300003	Computer Engineering
*	NULL	NULL	NULL	NULL	NULL	NULL

organizations

	organization_id	org_name	description	faculty_advisor_id
▶ 1	1	ACSES	Association of Computer Science & Engineering Students	2
2	2	CSI	Computer Society of India - SPIT Student Chapter	2
3	3	IEEE	IEEE SPIT Student Branch - Technical and research society	2
4	4	E - Cell	Entrepreneurship Cell SPIT - Startup incubation hub	2
5	5	Mudra (cultural)	Mudra - Cultural Committee overseeing performing arts	2
6	6	Sports committee	SPIT Sports Committee - Organizing sports tournaments	2
*	NULL	NULL	NULL	NULL

venues

	venue_id	venue_name	capacity
▶ 1	1	SPIT Auditorium	350
2	2	Seminar Hall 008	150
3	3	Central Computing Facility (CCF)	120
4	4	Computer Lab 202	60
5	5	College Quadrangle	500
*	NULL	NULL	NULL

memberships

	membership_id	organization_id	user_id	member_role
▶ 1	1	1	3	Chairperson
2	2	1	6	Secretary
3	3	1	4	Creatives Head
4	4	1	5	Marketing Head
5	5	2	5	Chairperson
6	6	4	4	Finance Secretary
*	NULL	NULL	NULL	NULL

events

	event_id	organization_id	venue_id	created_by	approved_by	event_name	event_date	max_participants	approval_status	event_status
▶1	1	1	3	3	1	CodeBash 2026	2026-10-25	100	Approved	Upcoming
2	2	1	4	4	1	Design & UI/UX Sprint	2026-09-15	50	Approved	Completed
3	3	1	2	5	1	Brand & Marketing Summit	2026-11-05	80	Approved	Upcoming
4	4	4	1	3	1	E-Summit: Shark Tank SPIT	2026-11-20	250	Approved	Upcoming
5	5	5	5	6	1	Tarang: Cultural Showcase	2026-09-22	400	Approved	Completed
6	6	6	5	5	NULL	Agility Cup: Football League	2026-12-02	150	Pending	Upcoming
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

registrations

	registration_id	event_id	user_id	participation_status
▶ 1	1	2	3	Attended
2	2	2	5	Attended
3	3	2	6	Attended
4	4	5	3	Attended
5	5	5	4	Attended
6	6	1	5	Registered
*	NULL	NULL	NULL	NULL

feedback

	feedback_id	registration_id	rating	comments
▶ 1	1	1	5	Great hands-on design sprint! Figma prototyping was explained thoroughly.
2	2	2	5	Exceptional creative branding session. Marketing materials insights were practical.
3	3	3	4	Very insightful technical workshop organized by ACSES.
4	4	4	5	Tarang was electrifying! Stellar musical battles and great lighting.
5	5	5	5	Flawless event coordination and vibrant performances by the cultural squad.
*	NULL	NULL	NULL	NULL

announcements

	announcement_id	organization_id	created_by	title	description
▶ 1	1	1	3	CodeBash 2026: Team Registrations Open	Registrations live on HackerRank! Teams of 2 to 3 before Oct 22.
2	2	1	4	ACSES Creatives & Design Submissions	Call for banner artwork, event posters, and teaser video edits.
3	3	1	5	ACSES Marketing & Sponsorship Drive	Applications invited for student marketing associates for annual symposium.
4	4	4	3	Shark Tank SPIT: Pitch Deck Deadline	Submit 5-slide startup pitch deck by Nov 15 for investor pitches.
5	5	5	6	Auditions for Mudra Dance Squad	Auditions for the annual intra-college dance troupe this Friday in Quadrangle.
6	6	6	5	Agility Cup: Fixture Schedule Released	Inter-department cricket and football draws published on sports noticeboard.
*	NULL	NULL	NULL	NULL	NULL

5. ER Diagram

